



Multimodal Deep Learning Search Engine for E-commerce Fashion

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Problem Definition

Many e-commerce sites still rely on TF-IDF / BM25-style keyword search (especially smaller retailers — Shopify defaults)

Large players use neural / hybrid semantic search, but the trade-offs aren't well documented for small to mid-sized catalogs

Question: When does a multimodal dual-encoder actually outperform keyword search?

We test both on the same dataset with the same metrics to find out



Dataset & Exploration

Fashion Product Images Dataset (Kaggle)

~44,400 product images + metadata (gender, color, category, season, usage)

Each product: images/{id}.jpg + styles.csv + styles/{id}.json

Split: 80% train, 10% val, 10% test

Built 4 caption styles per product: templated, brand, shopper, short



Methodology

Dual-encoder architecture (Tensorflow/Keras)

- Image encoder: EfficientNetB0 (ImageNet pretrained) → 384-dim
- Text encoder: sentence-transformers all-MiniLM-L6-v2 (frozen, pretrained)

Trained with contrastive learning

Baseline: TF-IDF keyword search on same product text

At inference: cosine similarity between query embedding and gallery



Evaluation Setup

Metrics: Top-1 Accuracy, Top-5 Accuracy, MRR, Precision@5

Gallery: 4,427 test products

Tested across 4 query styles to measure robustness:

- Templated: "A men black watches, for formal wear..."
- Brand: "Titan Men Black Watch"
- Shopper: "men's black formal watch"
- Short: "black watch"

Apples-to-apples: same metrics, same test set for both models



Results

Model	Query Style	Top-1	Top-5	MRR
TF-IDF	templated	0.52	0.81	0.65
TF-IDF	shopper	0.08	0.22	0.16
Dual-encoder	templated	0.17	0.46	0.31
Dual-encoder	shopper	0.07	0.24	0.17

TF-IDF dominates when query shares vocabulary with catalog

Dual-encoder is competitive on shopper-style natural language



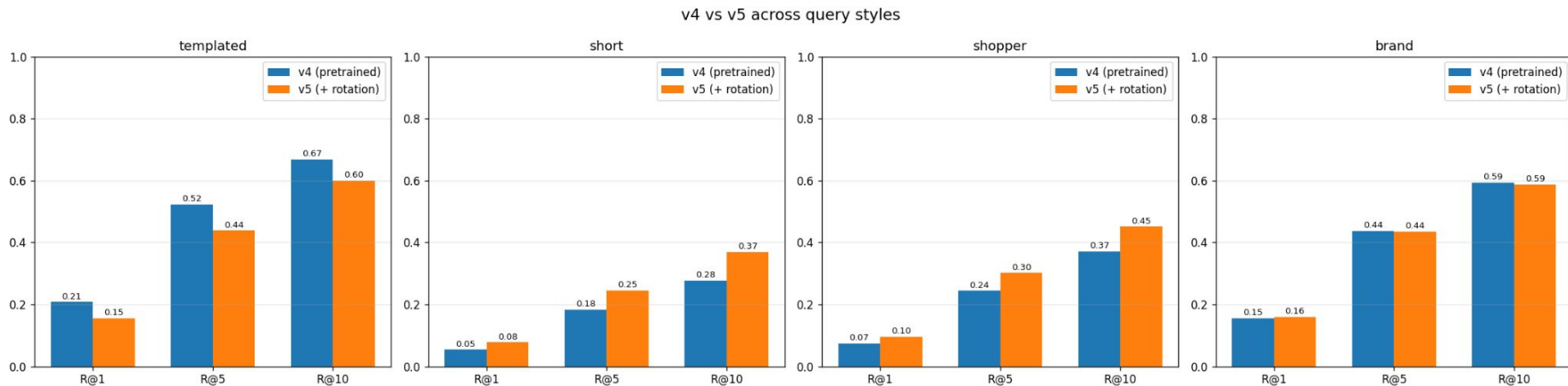
Interpretation & Iteration

Built 5 model versions (v1 → v5):

- v1: from scratch text encoder → hit a vocabulary ceiling
- v2-v3: caption augmentation → false negatives hurt performance
- v4: pre trained MiniLM → biggest single jump
- v5: + caption rotation → better balance, worse peak

Training a text encoder from scratch on 35k pairs can't compete with pretraining on 1B+ sentence pairs

v4 vs. v5 - Robustness Across Query Styles





Teamwork & Challenges

Division of work:

- Richardson: Dual-encoder model, training pipeline, v1-v5
- Samii: TF-IDF baseline, evaluation framework, docs

Practical challenges:

- Overfitting at epoch 5 - fixed with early stopping + `restore_best_weights`
- False-negative batches in caption expansion — fixed with rotation
- Apple Metal GPU compatibility — used legacy Adam optimizer

Baselines matter — a strong TF-IDF baseline reframed our whole story



Conclusion

Built complete multimodal retrieval pipeline + comprehensive baseline comparison

Keyword search remains very strong when the index is text-rich

Dual-encoders shine on out-of-vocabulary natural-language queries

Future work: Larger contrastive batches, paraphrase augmented training